

$$0.21 + 0.45 + 1.09 + 1.50 = 3.25 \text{ km}$$

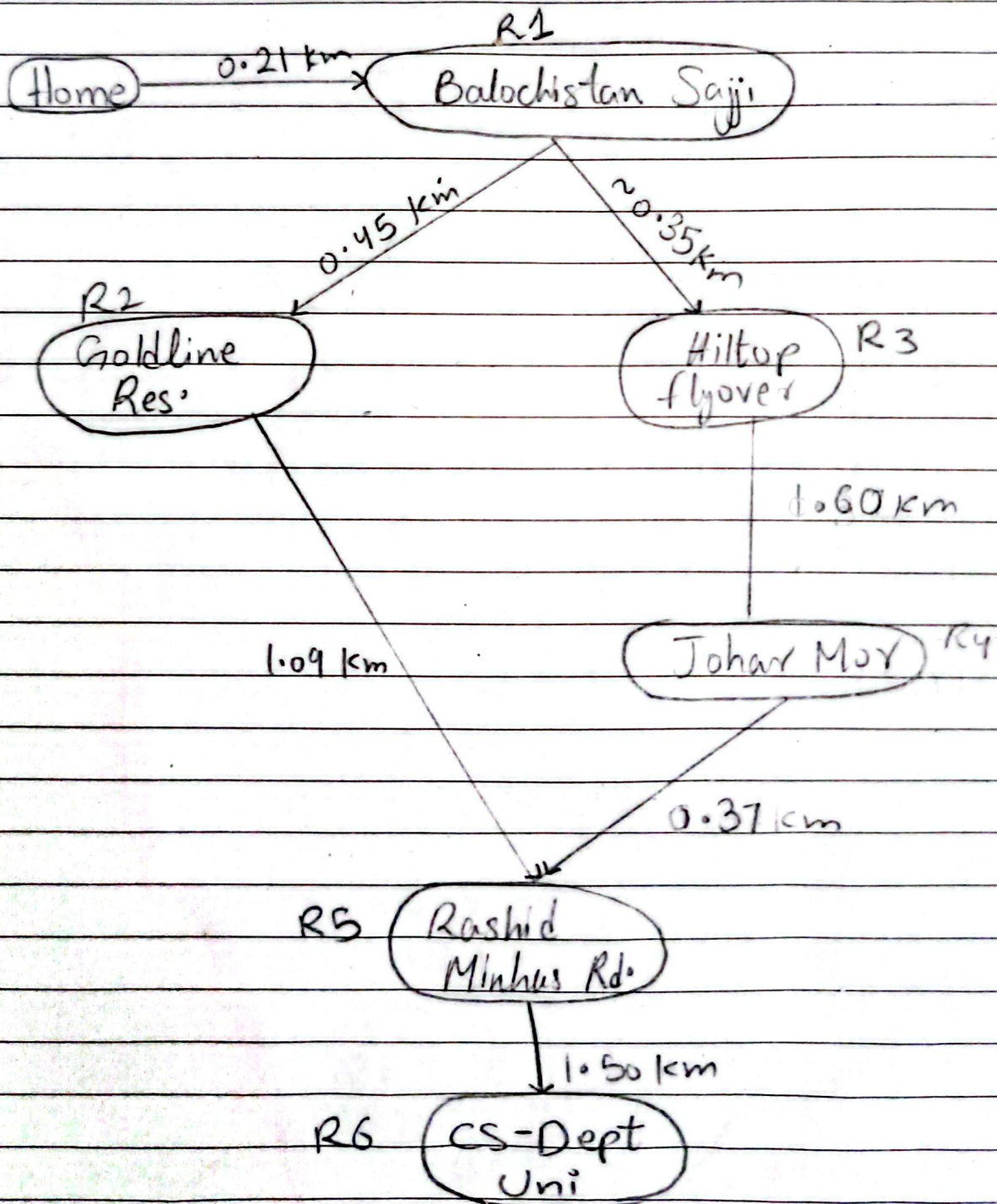
$$0.21 + 0.35 + 1.60 + 0.69 = 4.03 \text{ km}$$

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BFS

Home $\rightarrow$ Uni



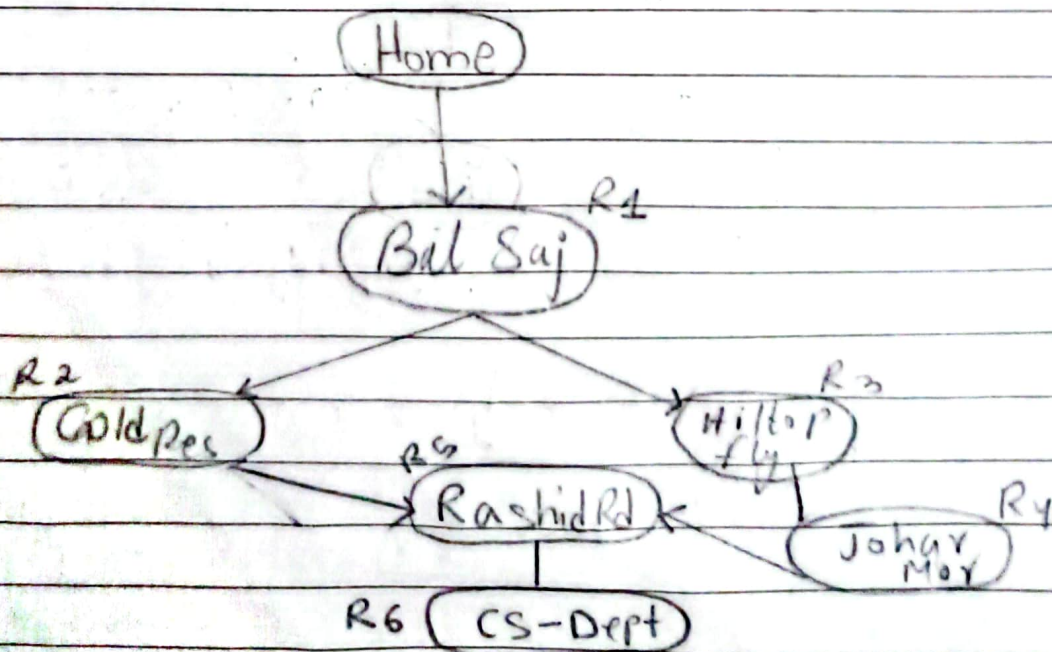
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Graph Representation

Node	Connects to
Home	R_1
R_1 (Bal Saj)	Home, R_2 , R_3
R_2 (Gold Res)	R_1 , R_5
R_3 (Hilltop fly)	R_1 , R_4
R_4 (Johar Mor)	R_3 , R_5
R_5 (Rashid Rd)	R_2 , R_4 , R_6
R_6 (Cs-Dept)	R_5

BFS-Tree:



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VVS®

Signature _____

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BFS - Trace

* S0: initialization

Q: [Home]

V: [Home]

* S1: Deq Home $\rightarrow$ Expand Home $\rightarrow$ (R1)

Q: [R1]

V: [Home, R1]

* S2: Deq R1 $\rightarrow$ Expand R1 $\rightarrow$ (Home, R2, R3)
Home already visited

Q: [R2, R3]

V: [Home, R1, R2, R3]

* S3: Deq R2 $\rightarrow$ Expand R2 $\rightarrow$ (R1, R5)
R1 already visited

Q: [R3, R5]

V: [" , R3, R5]

* S4: Deq R3 $\rightarrow$ Expand R3 $\rightarrow$ (R1, R4)
R1 already visited

Date: _____

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Q: [R₅, R₄]

V: [", R₃, R₄]

S₅: Deq R₅ → Expand R₅ → (R₂, R₄, R₆)
R₂ alr visited

Q: [R₄, R₆]

V: [", R₅, R₆]

S₆: Deq R₄ → Expand R₄ → (R₃, R₅)
R₃ alr visited

Q: [R₆]

S₇: Deq R₆ → Expand R₆ → (R₅)
R₅ alr visited

→ Goal founded

Summary:

→ Shortest Path (edges):

Home → R₁ → R₂ → R₅ → R₆ (4 edges)

→ BFS order of expansion:

Home → R₁ → R₂ → R₃ → R₅ → R₄ → R₆